

Springer Texts in Statistics

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An Introduction to Statistical Learning

with Applications in R

 Springer

ANALYTICS

Week 4-5

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*Examples Inspired By
CAS, SOA, Coaching
Actuaries ©*

Key Formulas This Week

- $\mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$, $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}$, $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$
- $Var(\mathbf{b}) = s^2(\mathbf{X}'\mathbf{X})^{-1}$
- $MSE = s^2 = \frac{1}{n-(p+1)} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{SSE}{n-(p+1)}$
- $R_a^2 = 1 - \frac{s^2}{s_y^2} = 1 - \frac{\frac{SSE}{n-(p+1)}}{\frac{SST}{n-1}} = 1 - \left(1 - R^2\right) \left(\frac{n-1}{n-(p+1)}\right)$

MLR Model

- The multiple linear regression model is:
- $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ or
 - $\mathbf{y}$: Response vector
 - $\mathbf{X}$: Design matrix
- $Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots \beta_p x_p + \varepsilon$
- $E[y] = \beta_0 + \beta_1 x_1 + \cdots \beta_p x_p$
- $\hat{y} = b_0 + b_1 x_1 + b_2 x_2 + \cdots b_p x_p$

Matrix Notation

- $Y = X\beta + \varepsilon$ is short-hand for:

$$\bullet \begin{bmatrix} Y_1 \\ Y_2 \\ \dots \\ Y_n \end{bmatrix} = \begin{bmatrix} 1 & x_{1,1} & \dots & x_{1,p} \\ 1 & x_{2,1} & \dots & x_{2,p} \\ \dots & \dots & \ddots & \dots \\ 1 & x_{n,1} & \dots & x_{n,p} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \dots \\ \beta_p \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \dots \\ \varepsilon_n \end{bmatrix}$$

Matrix Notation 2

- n : # of distinct data points
- p : # of predictor variables
- $\mathbf{X}$: $n \times p$ matrix of x'_{ij} s
- $\mathbf{X}^T$: Transposed matrix $p \times n$
- $\mathbf{x}_i$: vector of length p (1 row of $\mathbf{X}$)
- $\mathbf{x}_p$: vector of length n (1 column of $\mathbf{X}$)
- $\mathbf{y}$: vector of outputs of length n
- $(\mathbf{AB})_{ij} = \sum_{k=1}^d a_{ik} b_{kj}$ matrix multiplication

MLR Model 2

- $\hat{y} = b_0 + b_1x_1 + b_2x_2 + \cdots b_px_p$
- MLR only works if $n \geq p + 1$. When $n = p + 1$, the fit is _____
- Linear regression is considered a relatively inflexible model, though it is more flexible than it looks because MLR only requires the _____

Coefficient Estimation

- The $b_0, b_1, \dots, b_p$ are still estimated using ordinary least squares, by minimizing:
- $\sum_{i=1}^n (y_i - \hat{y}_i)^2$
- Under the method of least squares,
- $\mathbf{b} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$
 - $\mathbf{X}^T$: Transpose of $\mathbf{X}$

Coefficient Estimation 2

- $\hat{\mathbf{y}} = \mathbf{X}\mathbf{b} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} = \mathbf{H}\mathbf{y}$
- $\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ is known as the hat matrix

Coefficient Interpretation

- Each β_j represents the expected change in y per unit change in x_j , holding all other variables fixed.
- For logarithmic y , the % change is what is linear.
- Positive coefficients infer a
- Negative coefficients infer an inverse linear relationship with the response variable.

1. Estimation Illustration

- A multiple regression model used to estimate # points scored on Final is fitted to 20 observations. Predictors are Exam 1 Score X_1 , Interest Level X_2 , and # of hours of sleep X_3 .
- $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon$
- $(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = \begin{pmatrix} 1.87 \\ 4.44 \\ 1.6 \\ 0.22 \end{pmatrix}$
- $\mathbf{X}^T \mathbf{Y} = \begin{pmatrix} 0.8 \\ 1.2 \\ 1.5 \\ 0.1 \end{pmatrix}$, $x_* = \begin{pmatrix} 1 \text{ (Always)} \\ 0.8 \\ 0.9 \\ 7 \end{pmatrix}$
- Calculate the predicted value for the given inputs x_* .
- Interpret the b_1 coefficient.

1. Estimation Illustration 2

- $\mathbf{b} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$
- $\hat{\mathbf{y}} = \mathbf{X} \mathbf{b}$

Model Assumptions

- Multiple linear regression model assumptions:
- 1. $Y_i = \beta_0 + \beta_1 x_{i1} + \cdots \beta_p x_{ip} + \varepsilon_i$ is the model
- 2. $\{x_{i1}, x_{i2}, \dots, x_{ip}\}$ are **non-stochastic variables**
- 3. $E[\varepsilon_i] =$
- 4. **Homoscedasticity:**

Model Assumptions 2

- Multiple linear regression model assumptions:
- 5. y'_i s (or equivalently ε'_i s) are ____
- 6. y'_i s (or equivalently ε'_i s) are ____
- 7. Predictors are **non-collinear**

Least Squares Properties

- Gauss-Markov Theorem: If assumptions 1-5 hold, the least squares estimator $\mathbf{b}$ is the **minimum variance unbiased estimator** (MVUE) of $\boldsymbol{\beta}$, meaning it is the _____
- If assumptions 1-6 hold, then $\mathbf{b}$ is normally distributed.

Var-Cov Matrix

- $Var(\mathbf{b}) = s^2(\mathbf{X}'\mathbf{X})^{-1} =$

- $$\begin{pmatrix} Var(b_0) & Cov(b_0, b_1) & \dots & Cov(b_0, b_p) \\ Cov(b_1, b_0) & Var(b_1) & \dots & Cov(b_1, b_p) \\ \vdots & \vdots & \ddots & \vdots \\ Cov(b_p, b_0) & Cov(b_p, b_1) & \dots & Var(b_p) \end{pmatrix}$$

- This is the variance-covariance matrix.

Mean Square Error

- The mean square error s^2 is used to estimate the true unknown σ^2 (irreducible error).
- $MSE = s^2 = \frac{1}{n-(p+1)} \sum_{i=1}^n (y_i - \hat{y}_i)^2 =$
 - p : # of explanatory parameters ($b_1, b_2, \dots, b_p$)
 - $s = RSE = \sqrt{MSE}$: residual standard deviation/error

Example 1

- The monthly electricity consumption of 25 households was studied using MLR.
- The coefficient of determination is 0.80.
- The unbiased sample variance of the monthly electricity consumption is 400.
- Calculate the variance estimate of the monthly electricity consumption.

	Coefficient
Intercept	100
# of Appliances	15
High Travel	-20
Suburban	10

Example 1 Sol

- $MSE = s^2 = \frac{SSE}{n-(p+1)}$
- $s_y^2 = \frac{SST}{n-1}$
- $R^2 = \frac{SSReg}{SST}$
- $SST = SSE + SSReg$

Sum of Squares

- $SST =$
- $\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$
 - SSE : Sum of unexplained deviation squared
 - $SSReg$: Sum of explained deviation squared
- Coefficient of determination: $R^2 = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} =$

SLR Reminders

- In simple linear regression:
- $R^2 = r^2$
- $F = t^2$
- Least-squares line passes through $(\bar{x}, \bar{y})$

Adjusted R^2

- A drawback of R^2 is that it never decreases as more explanatory variables are added, regardless of if they are useful.
- Adjusted R^2 attempts to penalize using degrees of freedom and can help compare **nested models**, where _____

Adjusted R^2 2

- $R_a^2 = 1 - \frac{s^2}{s_y^2} =$

- Adjusted R^2 represents the proportional reduction in the residual variance of the model.
- Higher adjusted R^2 is preferable.

Adjusted R^2 Properties

- Adjusted $R^2 \leq R^2$ and only equals R^2 when _____
- Adjusted R^2 does not have to be _____
- If two models are nested, the larger model has higher R^2 , but we cannot make any conclusions of the adjusted R^2 .

ANOVA Table

- The **ANOVA** (analysis of variance) table summarizes the sources of variability.
- Mean square is the sum of square values divided by its respective degrees of freedom.

Source	Sum of Squares	df	Mean Square
Regression	$SSReg$	p	$MSR = \frac{SSReg}{p}$
Error	SSE	$n - (p + 1)$	$MSE = \frac{SSE}{n - (p + 1)}$
Total	SST	$n - 1$	

Concept Check #1

- Given models run on the same data, which statements are true?
- $R_A^2 = 0.6$ and Adjusted $R_A^2 = 0.55$
- $R_B^2 = 0.7$ and $R_C^2 = 0.5$
- I. Model B is preferred because it explains more of the response variability than A.
- II. It is impossible for Model B predictors to be a strictly smaller subset of Model A predictors.
- III. If Model D has $R^2 > 0.55$ and has fewer predictors than A, it is possible that A and D are nested.
- IV. Adjusted R_B^2 must exceed 0.55.
- V. Under Adjusted R^2 , Model A is preferable over C.

2. Adjusted R^2 Illustration

- Consider a dataset of 100 observations with sample standard deviation of the response of 100. We run a regression with 3 explanatory variables to get residual standard error of 50.
- Calculate the adjusted coefficient of determination.
- Complete the ANOVA table.
- Calculate the unadjusted coefficient of determination.

2. Adjusted R^2 Illustration 2

- $R_a^2 = 1 - \frac{s^2}{s_y^2} = 1 - \frac{\frac{SSE}{n-(p+1)}}{\frac{SST}{n-1}} = 1 - \left(1 - R^2\right) \left(\frac{n-1}{n-(p+1)}\right)$
- $SST = SSE + SSReg$

2. Adjusted R^2 Illustration 3

Source	Sum of Squares	df	Mean Square
Regression			
Error			
Total			

Example 2 (CAS)

- From an ordinary linear regression with 5 parameters (includes intercept) and 50 observations used to predict the success of a Korean pop song, you find that the total sum of squares is 996.
- A. If the coefficient of determination is 0.864, calculate the adjusted R^2 .
- B. If instead, the unbiased estimate for the constant error term variance is 2, calculate $1 - \frac{\sum_{i=1}^{50} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{50} (y_i - \bar{y})^2}$.

Example 2 Sol

- $R_a^2 = 1 - \frac{s^2}{s_y^2} = 1 - \frac{\frac{SSE}{n-(p+1)}}{\frac{SST}{n-1}} = 1 - \left(1 - R^2\right) \left(\frac{n-1}{n-(p+1)}\right)$
- $MSE = s^2 = \frac{SSE}{n-(p+1)}$
- $R^2 = 1 - \frac{SSE}{SST}$

Polynomial Regression

- Multiple linear regression can handle **non-linear** relationships because the explanatory variables can be transformed in any fashion.
- Polynomial regression: $\hat{y} = b_0 + b_1x + b_2x^2 + \dots b_px^p$
- For a p^{th} order polynomial, convention is to include a term for a powers from 1 to p , even if a model finds smaller orders insignificant.
- Disadvantage: Harder to interpret model

Transformed Variables

- Logarithmic regression: $\hat{y} = b_0 + b_1 \ln x$
- Exponential regression: $\hat{y} = b_0 + b_1 e^x$
- x can even be categorical.

Common Transformations

- Because the normal distribution has so many fantastic properties to work with, transforming **skewed** distributions to approximately normal can greatly help with modeling.
- Lognormal: $\ln x$
- Data w/0's: Use $\ln(1 + x)$
- Data w/negative %s: Use $100 \ln \left(1 + \frac{x}{100} \right)$
- Square Root: $\sqrt{x}$

Binary Variables

- **Categorical variables** measure observations into distinct groups/categories. They are _____
- **Binary** variables are categorical variables that can take only one of two values.
- For these variables, a **dummy**/indicator variable is used in the x_j (ie. 0 or 1 for binary).

Binary Variables 2

- In the base case for a binary variable ($x_j = 0$), the beta is not present.
- In a basic linear regression case with binary x_1 ,
- $$\hat{y} = \begin{cases} b_0 + b_1, & \text{for other } (x_1 = 1) \\ b_0, & \text{for base case} \end{cases}$$
- b_1 is the expected change on y when going from the

Categorical Variables

- For a categorical variable with m categories, you will need _____
- Choice of the **baseline** will affect coefficient estimates, but predictions always _____

Categorical Variables 2

- For example, for a $\{High, Medium, Low\}$ variable
- $x_{i1} = \begin{cases} 1, i^{th} \text{ is High} \\ 0, i^{th} \text{ is not High} \end{cases}, x_{i2} = \begin{cases} 1, i^{th} \text{ is Low} \\ 0, i^{th} \text{ is not Low} \end{cases}$
- $\hat{y} = b_0 + b_1x_1 + b_2x_2$
- Medium is the baseline and b_1 would be the average difference between High and Medium, and b_2 the average difference between Low and Medium.

Categorical Convention

- In practice, if you include a categorical variable, you should include all the dummy variables, even the non-significant ones, or else the interactions change.

Interaction Terms

- **Interaction terms** are added when explanatory variables have a non-linear (most commonly multiplicative) effect on the response. They account for _____
- ie. $\hat{y} = b_0 + b_1x_1 + b_2x_2 + b_3x_1x_2$

Interaction Terms 2

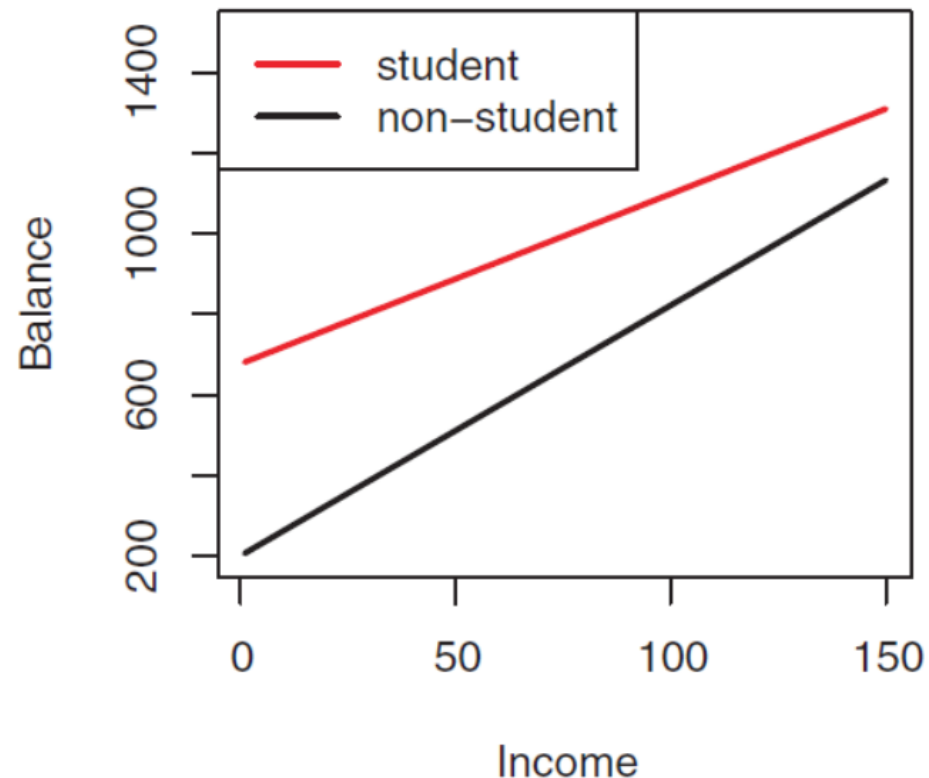
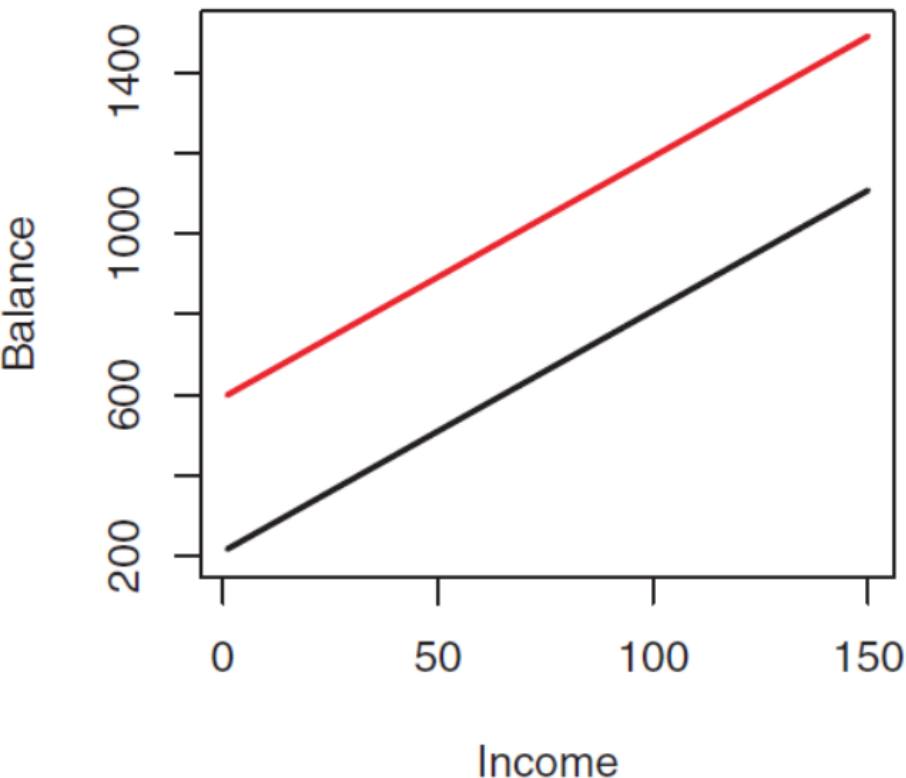
- $\hat{y} = b_0 + b_1x_1 + b_2x_2 + b_3x_1x_2$
- b_3 models the interaction between x_1 and x_2 in influencing y .

Categorical Interaction

- Binary Example: Let $X_2 = 1$ if male, 0 if female and X_1 be continuous
- $\hat{y} = b_0 + b_1X_1 + b_2X_2 + b_3X_1X_2$
- $\hat{y} = \left\{ \right.$

Interaction Terms (ISL)

- Without an interaction, the model gives both student/non- the same slope. With, the slopes differ



Interaction Terms 3

- **Hierarchical principle:** If an interaction is included, also include the main effects, even if the p -values associated with the coefficients are _____
- **Warning:** The interaction term must not be a linear combination of _____

Concept Poll #2 (© SOA)

- Which statements are true?
- I. Because the logarithm of zero is negative infinity, a logarithm transformation cannot be used if the data has 0s.
- II. A square root transformation may help make the variance of residuals more constant.
- III. If spread of the residuals increases as the predicted values increase, the data exhibits homoscedasticity.
- IV. If $Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$ with x_1, x_2 as dummy categorical variables, the data is separated into four distinct classes by the categorical variables.

3. Interaction Illustration

- Salary is predicted using two categorical variables and their interactions.
- GPA is a binary variable where Above3 is the reference category.
- Exams is a categorical variable with 3 levels, where 1 is the reference category.
- Coefficient estimates after least squares:
 - Intercept = 80, Below3 = -10, Exams0 = -5, ExamsAbove1 = 6, Below3:Exams0 = -7, Below3:ExamsAbove1 = 11
- All else equal, what is the impact on salary of:
 - A. 0 exams vs. Above 1 exams for Above 3.0 GPA?
 - B. A GPA below 3.0 (directionally for each case)?
 - C. What's the intercept of the fitted linear equation for 0 exams individuals?

3. Interaction Illustration 2

GLM Notation

- On HW, you will see some problems that tell you the Response Distribution is Normal and Link is Identity.
- That's a fancy way of saying multiple linear regression and will be covered in GLMs later.

Example 3 (CAS)

- You fit a linear model using the following two-level categorical variables:
- $X_1 = 1$ if Female and 0 if Male
- $X_2 = 1$ if Multi-Car and 0 if Single-Car
- Y represents # of accidents.
- Actuary A uses: $E[Y] = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2$ and finds
- $b_0 = 0.75, b_1 = -0.5, b_2 = 1, b_3 = -0.25$
- Actuary B defines $Y_1 = 1$ if Male and 0 if Female, $Y_2 = 1$ if Single-Car and 0 if Multi-Car with the same data.
- Find the coefficients of Actuary B's model.

Example 3 Sol

- The predicted values are the same regardless of variable coding.

Example 4

- You use the following data to estimate # of kids (y) using a linear regression model with an intercept, a continuous predictor income (x_1), a binary predictor homeownership (x_2) where Yes = 0 is the base case, and an interaction term between the two variables.
- Determine the coefficient of determination of the model, each residual, and the estimated coefficient for income.

y	Income	Homeowner
5	2	Yes
1	0	Yes
1	1	No
0	2	No

Example 4 Sol

- If $n = p + 1$, the MLR fit is perfect.

Summary Check

- *Can you describe in your own words:*
- Multiple linear regression assumptions
- Recognize the $\mathbf{b}$, $\mathbf{H}$, $\hat{\mathbf{y}}$, variance-covariance matrices
- R^2 vs. Adjusted R^2 properties
- Formulas for MSE , R^2 , Adjusted R^2
- How dummy variables, interaction terms work